

Flux: AI-Powered PCB Design Assistant

Flux is your AI-powered design assistant that lives in your Flux Editor projects, answers complex questions, and helps you move faster by automating PCB design tasks.

Overview

Flux is an AI-powered hardware design assistant integrated directly into Flux Editor. This custom-trained large language model (LLM) understands your schematics, components, electrical connections, and bill of materials. **With Flux, you can accelerate part selection, evaluate alternatives, receive design feedback, and with your approval, make direct changes to your schematic.**

Join the Flux channel in our [Slack community](#) to share your experiences and discover new use cases.

Getting Started with Flux

Access Flux through the Chat tab in the right sidebar or by right-clicking anywhere on your project canvas and selecting "Flux". For detailed guidance on using AI in your design process, see our [AI-assisted PCB design tutorial](#).

Starting a conversation

Simply type your question or request in the Chat tab. Flux maintains context within conversation threads, allowing for natural back-and-forth exchanges about your design.

Info

Flux may take a few seconds to respond as it processes your design information.

What can Flux do?

- **Answer questions:** Get relevant information for part selection, design feedback, and circuit optimization. [Learn more](#)
- **Parse datasheets:** Save time by having Flux extract and reference datasheet information. [Learn more](#)

- **Modify your design:** Let Flux make connections and changes directly in your schematic with your approval. [Learn more](#)
- **Access documentation:** Get help with Flux Editor features and workflows. [Learn more](#)
- **Find components:** Search the library for parts that meet your specifications. [Learn more](#)
- **Generate FMEA reports:** Identify potential failure modes and assess risks with [FMEA analysis](#)
- **Perform calculations:** Evaluate engineering formulas and analyze circuit parameters. [Learn more](#)
- **Generate code:** Create Python scripts for data analysis and visualization. [Learn more](#)
- **Simulate circuits:** Verify circuit behavior using SPICE simulation. [Learn more](#)

Specialized Tools

- [@file](#) – Access attached datasheets and project files
- [@library](#) – Search the Flux component library for part selection
- [@calculator](#) – Extract equations from datasheets for calculations
- [@code](#) – Generate Python scripts and charts
- [@help](#) – Search Flux documentation for guidance
- [@simulator](#) – Perform SPICE circuit simulations

Project Context Access

Flux analyzes your project to provide relevant suggestions:

- **Project metadata:** Uses project name and description for context
- **Components and netlist:** Understands your circuit's structure and connections
- **Part information:** Accesses datasheets for compatibility and design recommendations
- **Layout:** Understands your projects layout with its footprints, silk and traces
- **Comment locations:** Recognizes the physical context of your questions
- **Flux Editor documentation:** Provides guidance on tool features and workflows

Info

Flux primarily works with schematics and currently has limited understanding of PCB layout and trace positioning.

Key Features

Chat Interface

- **Model Selection:** Choose AI models optimized for different tasks
- **Context Controls:** Select what project information Flux uses
- **File Attachments:** Upload datasheets and reference documents
- **Branching Conversations:** Explore multiple design approaches
- **Response Regeneration:** Get alternative suggestions with "Try Again"

Design Actions

Flux can directly modify your design upon request:

- **Component Management:** Add, edit, or remove components
- **Designator Changes:** Rename and organize component references
- **Net Modifications:** Create and modify electrical connections
- **Property Editing:** Update component and design parameters

Keyboard Shortcuts

Action	Shortcut
Open Chat Tab	Alt + C
Focus on chat input	/
Submit message	Ctrl + Enter
Navigate history	Up/Down arrows

For detailed usage instructions, see our [Getting Started with Flux](#) tutorial.

Privacy and Security

Your design data remains private and secure. For details on our privacy practices, see our [Privacy Statement](#).

Current Limitations

- **Thread-specific context:** Flux only considers messages within the current conversation thread

Troubleshooting

Issue	Solution
No response	Ensure you're using the Chat tab or right-click menu and have sufficient ACUs
Incorrect actions	Review proposed changes before approval and use "Try Again" for alternatives
Context confusion	Break complex questions into smaller queries and provide specific context

ACU System

Flux uses ACUs (Agent Compute Units) for AI usage beyond free limits. **All plans include monthly ACUs**, with additional ACUs available on Pro and Organization plans. Learn more about [individual ACUs](#) or [organization ACUs](#).

Next Steps

- [AI for Hardware Design](#) - Leverage AI in your design process
- [Flux Use Cases](#) - Practical applications
- [Getting Started with Flux](#) - Step-by-step guide
- [Schematic Editor](#) - Master the design environment